

Prototype Shopping List

Steel Beam Cutting Cell - 1:10 desk-top prototype | Ravi Mahadeva | 05 October 2026

**TOTAL BUYING COST: about
£1,140**

incl. VAT where known, before postage | with 10% spare for postage and price changes: **about £1,260** | to get moving first (stages 1-2): about £430

Stage	About
Stage 1 - Frame and bed	£130
Stage 2 - Motion and controller	£300
Stage 3 - Arms and tools	£100
Stage 4 - Cameras and sensors	£130
Stage 5 - Safety and enclosure	£360
Stage 6 - Electrical	£90
Consumables	£40
Total	about £1,140

Before you buy

- Prices are approximate (GBP, October 2026) and change - check each one before ordering.
- **Every item has a link** (blue): a product page where it was checked, otherwise that shop's search.
- **Part numbers** (P01 ...) are the same as the balloons in docs/Prototype_Assembly.pdf, the Prototype tab in the app and the CAD - so you can see where each part goes.
- You already have the **Jetson Orin Nano**; it is not in the total.
- Order the aluminium **cut to size** (lengths on the last page) - it saves a lot of work.
- Buy and wire the **safety parts (stage 5) before any motor moves**: the E-stop and safety relay must cut motor power. Have the mains wiring checked by a competent person.
- Cheapest way to start: stages 1 and 2 first (frame and moving gantry, about £430), then arms, cameras and the rest.
- A used Pilz PNOZ s3 is about £75 instead of £180 - but don't skip the safety relay or the E-stop.
- No real cutting on the prototype: the Cutter holds a **pen** that marks the cut lines.

Stage 1 - Frame and bed

OK	Part	Item	What to look for	Qty	Each	Line total	Where to buy (click)
	P01	V-slot 20x20 aluminium extrusion	cut to size - lengths in cad/prototype_cut_list.csv (frame, bed rails, bridges, Z axes, camera post) the whole frame is 20x20; lengths come from the CAD model	9.5 m	£8	£76	Ooznest (cut to size), Amazon UK Search ooznest.co.uk
	P02	Corner brackets + M5 T-nuts + M5 bolts	30 cast 20-series corner brackets, 120 drop-in T-nuts, M5x8 button heads	1 kit	£30	£30	Ooznest, Amazon UK Search ooznest.co.uk
	P03	Rubber feet	stick-on rubber feet 20 mm stops it sliding on the desk	4	£1	£4	Amazon UK Search amazon.co.uk
	P04	Bed rollers	10 mm printed sleeves on 3 mm silver steel rod (50 mm), 2 x 623ZZ bearings each print the sleeves; buy rod and bearings	6	£2	£12	Amazon UK, Simply Bearings Search amazon.co.uk
	P05	Roller brackets (3D printed)	PETG, clip on the 20x20 bed rails filament counted under Consumables	12	£0	£0	print 3D print / make it
	P06	Bed riser blocks (3D printed)	PETG blocks that hold the bed rails 60 mm above the base filament counted under Consumables	6	£0	£0	print 3D print / make it
	P07	Outfeed table board	3 mm plywood strip 700 x 90 mm	1	£4	£4	builders' merchant, Amazon UK Search amazon.co.uk
	P08	Outfeed table risers (3D printed)	PETG filament counted under Consumables	3	£0	£0	print 3D print / make it
	P09	Scrap tray (3D printed)	700 x 44 mm tray under the bed filament counted under Consumables	1	£0	£0	print 3D print / make it
Stage total						£126	

Stage 2 - Motion and controller

OK	Part	Item	What to look for	Qty	Each	Line total	Where to buy (click)
	P10	Mini V-wheel gantry plate kits	20-series mini gantry plate + 4 mini V-wheels (625 bearings) + eccentric spacers 2 per bridge end x 2 bridges + 1 per carriage x 2	6	£12	£72	Ooznest Search ooznest.co.uk
	P11	NEMA 17 stepper motor	1.5-2 A, 40 mm body, 5 mm shaft (a NEMA 14 is lighter if you prefer) X (one per bridge) and Y (one per carriage)	4	£12	£48	Ooznest, StepperOnline Search omc-stepperonline.com
	P12	NEMA 17 with integrated T8 lead screw	T8 x 2 mm lead, 150 mm screw, with brass nut Z axes	2	£22	£44	StepperOnline, Amazon UK Search omc-stepperonline.com
	P13	Z motor bracket (3D printed)	PETG shelf on top of the carriage plate, holds the Z and Y motors filament counted under Consumables	2	£0	£0	print 3D print / make it
	P14	Lead-screw nut block (3D printed)	PETG block with the brass T8 nut, fixed to the Z axis filament counted under Consumables	2	£0	£0	print 3D print / make it
	P15	GT2 belt 6 mm	5 m (2 per bridge along the rails, 1 per carriage)	1	£8	£8	Ooznest Search ooznest.co.uk
	P16	GT2 16T pulleys + idlers	4 x 16T 5 mm bore (cross shafts), 2 x 16T (Y), 8 idlers	1 set	£20	£20	Ooznest, Amazon UK Search ooznest.co.uk
	P17	5 mm cross shafts + bearings	2 x 360 mm 5 mm silver steel, 4 x 625ZZ in printed blocks, 2 x 5-5 mm couplers one motor drives both rails of a bridge	1 set	£12	£12	Amazon UK, Simply Bearings Search amazon.co.uk
	P18	Home / limit switches	mechanical microswitch with lever, NC wiring one per axis + 2 spare	8	£1.50	£12	Amazon UK Search amazon.co.uk
	P19	FluidNC 6-axis CNC controller (ESP32)	6 driver sockets, USB to the Jetson (e.g. Elecrow / Bart Dring 6x CNC Controller) flash once from the web installer - no programming	1	£40	£40	Elecrow, Tindie Open product page
	P20	TMC2209 stepper driver modules	StepStick format quiet; one per axis	6	£5	£30	Amazon UK, Ooznest Search amazon.co.uk
	P21	Motor cable + drag chain	5 m 4-core 0.25 mm2, 1 m 7x7 drag chain	1 set	£15	£15	Amazon UK Search amazon.co.uk
Stage total						£301	

Stage 3 - Arms and tools

OK	Part	Item	What to look for	Qty	Each	Line total	Where to buy (click)
	P22	MeArm-type 4-servo arm kit	small 4-servo arm with MG90S metal-gear micro servos (reach ~150 mm); swap the gripper for the pen / magnet about £37 each (MeArm pocket-sized kit)	2	£37	£74	Cool Components, Kitronik Open product page
	P23	Arm mount plate (3D printed)	PETG plate: bolts the arm base under the Z axis filament counted under Consumables	2	£0	£0	print 3D print / make it
	P24	PCA9685 16-channel servo board	I2C - wired straight to the Jetson's 40-pin header (pins 3 / 5), drives all 8 servos no extra controller needed for the arms	1	£6	£6	Amazon UK, kunkune Open product page
	P25	5 V power supply for the servos	Mean Well LRS-35-5 (5 V 7 A) - servos and magnet	1	£12	£12	RS, Farnell Search rs-online.com
	P26	Cutter 'torch' for the prototype (pen holder)	fine-liner pen in a sprung 3D-printed holder (marks every cut line on the model beam) no real cutting on the prototype: see docs/PROTOTYPE.md	1	£2	£2	print 3D print / make it
	P27	Handler magnet	5 V 20 mm lifting electromagnet (~2.5 kg hold) + MOSFET module + flyback diode switch it from a FluidNC output	1	£8	£8	Amazon UK Search amazon.co.uk
	P28	Model beams (3D printed)	UB 305x165 at 1:10 (30 x 16 mm), 500 mm long, with steel tape on the top flange the magnet needs steel to grip; filament counted under Consumables	4	£0	£0	print 3D print / make it
Stage total						£102	

Stage 4 - Cameras and sensors

OK	Part	Item	What to look for	Qty	Each	Line total	Where to buy (click)
	P29	USB wide-angle camera (safety zone)	1080p UVC, 90-120 deg view (e.g. Logitech C920/C922 or an Arducam USB wide-angle) plug and play on the Jetson; YOLO watches the cell	1	£50	£50	Amazon UK, Pimoroni Search amazon.co.uk
	P30	CSI camera (close-up of the Cutter)	Arducam IMX219 for Jetson Orin Nano (22-pin cable), wide-angle lens optional; checks the marked lines	1	£25	£25	Arducam, RobotShop UK Open product page
	P31	Powered USB 3 hub	4-port, own power supply controller + camera (+ spare)	1	£20	£20	Amazon UK Search amazon.co.uk
	P48	VL53L1X time-of-flight distance sensor	I2C breakout, up to 4 m - the datum sensor: finds where the model beam starts on the Jetson's I2C with the PCA9685	1	£12	£12	Pimoroni, The Pi Hut Search shop.pimoroni.com
	P49	Pen touch-off micro-switch	micro-switch with roller lever, on the sprung pen holder - finds the beam surface (the prototype's initial height sensing) one spare	2	£1.50	£3	Amazon UK, CPC Search amazon.co.uk
	P50	Load cell 1 kg + HX711	bar load cell 1 kg with HX711 amplifier - between the Handler's wrist and magnet: weighs the part (slipping, wrong part, not cut free)	1	£7	£7	Amazon UK, The Pi Hut Search amazon.co.uk
	P51	INA219 current sensor	I2C current sensor in the magnet supply - checks the magnet is really on not drawn (wiring, inside the box)	1	£5	£5	Pimoroni, Amazon UK Search amazon.co.uk
	P52	IR flame sensor module	IR flame sensor with digital output - demonstrates the fire detector NOT a certified fire detector - demo only	1	£3	£3	Amazon UK Search amazon.co.uk
	P53	Line laser module (Class 2)	650 nm line laser, Class 1 or 2 (1 mW or less) - with camera 2 it shows the beam's real profile never look into the beam; Class 2 or below only	1	£6	£6	Amazon UK Search amazon.co.uk
Stage total						£131	

Stage 5 - Safety and enclosure

OK	Part	Item	What to look for	Qty	Each	Line total	Where to buy (click)
	P32	Emergency stop station	yellow box, red 40 mm mushroom, turn to release, 2 NC contacts (e.g. Schneider Harmony XALK178 family) BS EN ISO 13850; NC contacts to the safety relay - add an NO/NC block for the Jetson	1	£42	£42	RS, Farnell, TME Search rs-online.com
	P33	Safety relay	Pilz PNOZ s3 24 V DC (dual channel, manual reset) or equivalent cuts the 24 V motor and 5 V servo supplies - this is how a real machine does it	1	£180	£180	RS, Farnell (new) / ~£75 used Search farnell.com
	P34	Door interlock switch	non-contact coded magnetic safety switch, NC (e.g. Pilz PSEN or Schmersal BNS) BS EN ISO 14119; wired into the safety relay	1	£60	£60	RS, Farnell Search rs-online.com
	P35	Reset push button	22 mm blue illuminated, 1 NO outside the enclosure with a view inside	1	£8	£8	RS, Amazon UK Search rs-online.com
	P36	Stack light	24 V mini LED tower, red / amber / green driven by the Jetson through a relay module	1	£15	£15	Amazon UK Search amazon.co.uk
	P37	4-channel relay module (opto-isolated)	3.3 V logic input, 24 V contacts stack light and magnet	1	£8	£8	Amazon UK Search amazon.co.uk
	P38	IR break-beam sensor pair	5 mm IR break-beam (demonstrates the light-curtain input) one pair demonstrates the light curtain, one is the bar-present photo-eye - NOT a safety light curtain	2	£6	£12	Pimoroni, The Pi Hut Search thepihut.com
	P39	Polycarbonate sheet 2 mm	front door, ends and lid (~0.6 m2) - polycarbonate, not acrylic (acrylic shatters)	0.6 m2	£35	£21	plastics stockist, Amazon UK Search amazon.co.uk
	P40	Hinges + handle	2 small hinges, 1 handle (20-series)	1 set	£12	£12	Ooznest, Amazon UK Search ooznest.co.uk
Stage total						£358	

Stage 6 - Electrical

OK	Part	Item	What to look for	Qty	Each	Line total	Where to buy (click)
	P41	24 V power supply	Mean Well LRS-100-24 (24 V 4.5 A) - motors, controller, safety circuit	1	£14	£14	RS, Farnell Search rs-online.com
	P42	IEC mains inlet with switch and fuse	panel mount, 3 A fuse have the mains side checked by a competent person	1	£8	£8	RS, Farnell Search rs-online.com
	P43	DIN rail + terminals + fuse holders	short 35 mm DIN rail, 12 terminals, 3 x 5x20 fuse holders	1 set	£20	£20	RS, Farnell Search rs-online.com
	P44	Control box	ABS enclosure ~200 x 160 x 110 with cable glands holds the Jetson too	1	£20	£20	RS, CPC Search rs-online.com
	P45	Wire + ferrules + crimp tool	0.5 and 0.75 mm ² flexible, ferrule kit	1 set	£25	£25	Amazon UK, CPC Search amazon.co.uk
	P46	Jetson Orin Nano Super (4 GB)	you already have it - docs/JETSON_SPECS.md runs the app, YOLO, the planner and the arms	1	£0	£0	- <i>3D print / make it</i>
Stage total						£87	

Consumables

OK	Part	Item	What to look for	Qty	Each	Line total	Where to buy (click)
	P47	PETG and PLA filament	1 kg each brackets, risers, tray, model beams, arm parts	2	£20	£40	Amazon UK, 3DJake Search amazon.co.uk
Stage total						£40	

Aluminium cut list (order cut to size)

OK	Profile	Length (mm)	Quantity	Used for
	20x20 V-slot	200	2	Z axis
	20x20 V-slot	280	4	cross member
	20x20 V-slot	300	6	upright
	20x20 V-slot	340	2	bridge
	20x20 V-slot	390	1	camera post
	20x20 V-slot	700	2	bed rail
	20x20 V-slot	900	2	base rail
	20x20 V-slot	900	2	top rail

The lengths come from the CAD model (cad/prototype_1to10.step). The build steps, the safety wiring and why each part was chosen are in docs/PROTOTYPE.md.